Fiscal Transfers and Local Financial Intermediation: Evidence from Mining Revenues in Peru

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Abstract

Do fiscal transfers deepen local financial systems? I study Peru's *canon minero*—the country's largest intergovernmental transfer—using a panel of up to 1,893 Peruvian districts (1,837 in the main estimation sample) over 2012–2024. A Bartik instrument interacting pre-period fiscal dependence with commodity prices provides exogenous variation in municipal spending. The reduced form shows that higher Bartik exposure significantly lowers local deposits ($p = 0.014$). IV estimates confirm a robustly negative effect of spending on deposits across all specifications and functional-form transformations. Restricting the sample to districts without mines preserves the result, ruling out direct mining exposure as a confounder. Meanwhile, mining activity itself has a positive effect on deposits. The contrast suggests that fiscal intermediation of resource revenues—channeled through centralized procurement—bypasses local banks, while private-sector mining activity does not. These findings contribute to understanding the financial dimension of the resource curse.

Keywords: fiscal transfers; financial intermediation; resource curse; Bartik instrument; Peru

JEL: H72, G21, O16, Q32

1 Introduction

Financial development is among the most robust predictors of economic growth (King and Levine, 1993; Levine et al., 2000; Jayaratne and Strahan, 1996). At the subnational level, the presence of bank branches has been linked to poverty reduction (Burgess and Pande, 2005), increased non-agricultural output, and improved access to credit. A separate literature documents that intergovernmental fiscal transfers can improve local public goods (Litschig and Morrison, 2013), reduce poverty (Zambrano et al., 2014), and raise economic activity, with estimated local fiscal multipliers in the range of 1.5–2.0 (Nakamura and Steinsson, 2014; Corbi et al., 2019; Chodorow-Reich, 2019). If transfers stimulate local economic activity, and economic activity generates demand for financial services, one might expect fiscal transfers to deepen local financial systems.

This paper tests this hypothesis and finds that it fails. Using Peru's *canon minero*—the country's largest intergovernmental transfer, funded by mining tax revenues—I show that exogenous increases in municipal spending *reduce* rather than increase local deposits. The reduced form relationship between the Bartik instrument and deposits is negative and significant (-0.61, $p = 0.014$), and this sign is robust across all IV specifications, sample restrictions, and functional-form transformations. There is also suggestive evidence of a negative effect on credit, though this result is less precisely estimated.

Why might fiscal transfers fail to generate—or even crowd out—local financial intermediation? Three channels are possible. First, *fiscal leakage*: if transfer-funded spending flows through centralized treasury systems and pays contractors based outside the district, the money may enter and leave without passing through local banks. Second, *crowding out of private activity*: if public investment substitutes for private economic transactions that would otherwise generate deposits and credit demand, the net effect on intermediation could be negative (Friedman, 1978; Boehm, 2020). Third, *institutional displacement*: governments funded by non-tax revenues may have weak incentives to develop financial infrastructure, as they do not depend on a local tax base that requires a functioning banking system (Menaldo, 2016).

I exploit Peru's canon minero system for identification. The canon distributes 50 percent of corporate income taxes from mining to subnational governments in mining-producing regions. These transfers are large—a median 43 percent of municipal budgets in recipient districts—and earmarked exclusively for public investment. Their magnitude fluctuates sharply with global commodity prices, providing substantial time-series variation in municipal spending that is plausibly exogenous to local financial conditions.

The identification strategy uses a Bartik (shift-share) instrument (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022): the interaction of each district's pre-period canon share (measured in 2012) with a national mining commodity price index. The first stage is strong ($F = 34.9$). Critically, the reduced form—regressing deposits directly on the instrument—yields a coefficient of -0.61 ($p = 0.014$), providing model-free evidence that higher fiscal exposure to commodity prices reduces local deposits.

A key threat to identification is that commodity price booms may affect local financial markets directly—through mining employment and wages—rather than through the fiscal channel alone. I address this concern by (i) controlling for direct mining exposure and (ii) restricting the sample to districts without mining activity. In both cases, the negative effect on deposits is preserved or strengthened, while direct mining exposure has a *positive* effect on deposits ($+0.34$, $p < 0.05$). Mining activity generates local deposits; mining-funded fiscal transfers do not.

This paper contributes to several literatures. First, it adds to research on the subnational effects of resource windfalls (Caselli and Michaels, 2013; Aragón and Rud, 2013; Loayza et al., 2016; Allcott and Keniston, 2018; Cust and Poelhekke, 2015). While prior work examines effects on income, employment, public goods, and conflict, I focus on financial intermediation—a channel that has received limited attention despite its importance for growth. Second, the paper provides microeconomic evidence consistent with cross-country findings that natural resource revenues hinder financial development (Bhattacharyya and

Hodler, 2014; Nili and Rastad, 2007; van der Ploeg, 2011). Third, it contributes to the literature on fiscal transfer multipliers (Corbi et al., 2019; Nakamura and Steinsson, 2014) by documenting a channel through which transfers may have unintended negative spillovers. Fourth, the paper complements the contemporaneous work of Contreras and Orihuela (2026), who document that mining transfers weaken municipal property tax collection in Peru. Taken together, these findings suggest that the subnational fiscal resource curse operates along multiple dimensions simultaneously: canon revenues reduce both own-source fiscal capacity and local financial intermediation. A companion study (Author, 2026) shows that financial access moderates the local resource curse in Peru; here, I show the other side: fiscal transfers do not build the financial infrastructure that would make them effective.

2 Institutional Background

2.1 The Canon Minero

Peru's *canon minero* is the largest intergovernmental fiscal transfer in the country, distributing 50 percent of corporate income taxes paid by mining companies to subnational governments in the regions where mining takes place.[1] The transfer follows a formula that allocates revenues across departments, provinces, and districts based on the location of extraction and population. The canon has been the subject of extensive research on local governance, conflict, and development (Arellano-Yanguas, 2011; Maldonado and Ardanaz, 2023).

Three features of the canon are important for this study. First, the transfers are *large*: in the median recipient district, canon revenues constitute 43 percent of the total municipal budget (interquartile range: 24–64 percent). Second, they are *volatile*: because they derive from corporate income taxes on mining, their magnitude fluctuates sharply with global commodity prices. A mining price boom in 2011–2013, followed by a decline through 2016

[1] The legal framework is established by Law 27506 (2001) and its subsequent modifications.

and a recovery after 2017, generated substantial within-district variation in fiscal resources. Third, canon revenues are *earmarked*: they must be spent on public investment (capital expenditure), not current spending or transfers to households. This means the fiscal channel operates through procurement, infrastructure, and public works—activities that may or may not involve local financial institutions.

2.2 Local Financial Intermediation in Peru

Peru's financial system is concentrated in urban areas. The Superintendencia de Banca, Seguros y AFP (SBS) reports that as of 2024, only 25 percent of Peru's roughly 1,900 districts have any formal financial institution—a commercial bank branch, municipal savings bank (*caja municipal*), rural savings bank, or branch of the state-owned Banco de la Nación. In rural and mining districts, financial coverage is even thinner.

Municipal governments interact with the financial system primarily through the national treasury system (SIAF-MEF). Canon revenues are deposited in government accounts at the Banco de la Nación, and spending is executed through the SIAF. Payments to contractors and suppliers can be made through the national payments system without involving local banks. This institutional structure means that fiscal resources can flow into and out of a district's government budget without generating local deposits or credit.

Figure 1 illustrates the identifying variation. Mining commodity prices fell sharply from 2012 to 2015, recovered through 2021, and stabilized thereafter. Municipal spending tracks this pattern with a lag, reflecting the fiscal dependence of canon-receiving districts on commodity prices.

3 Data

I construct a district-level panel covering up to 1,893 Peruvian districts over 2012–2024 (13 years). The theoretical panel has 24,590 district-year observations; the OLS estimation

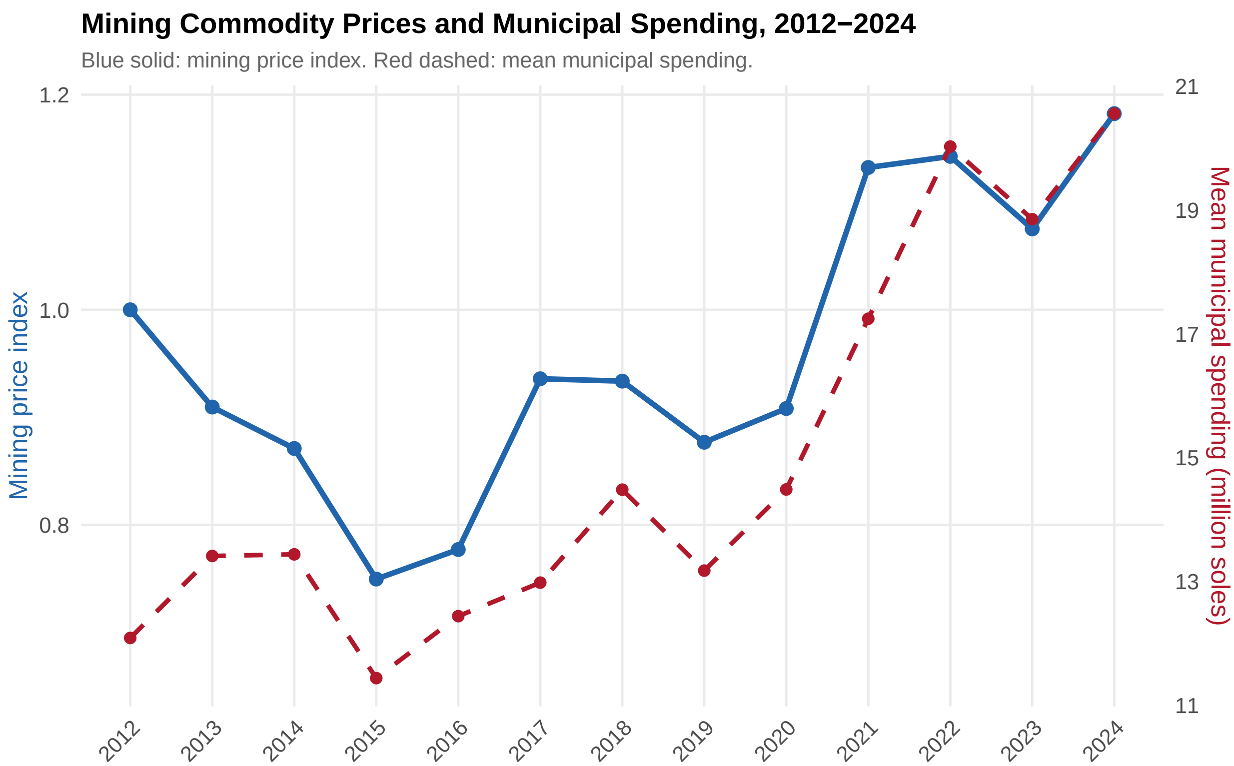

Figure 1: Mining Commodity Prices and Municipal Spending, 2012–2024

Notes: Mining price index (solid blue, left axis) is a national commodity price index normalized to 1.0 in the base year. Mean municipal spending (dashed red, right axis) is the average *devengado* across districts with valid canon share, in millions of soles.

sample is 24,304 (some districts missing spending data in certain years), and the IV sample is 23,862 (restricted to districts with a valid 2012 canon share, $N = 1{,}837$). The sample begins in 2012, the first year of the VIIRS nighttime lights satellite, and ends in 2024, the latest year with complete fiscal and financial data.

Municipal spending. Annual expenditure data come from the Consulta Amigable portal of the SIAF-MEF, Peru's public expenditure tracking system. I use total *devengado* (accrued expenditure) as the measure of municipal spending, and the *presupuesto institucional modificado* (PIM, modified budget) for calculating the canon share. Median annual spending is 5.7 million soles (Table 1).

Canon revenues. Mining canon revenues at the district level are also from the SIAF. To construct the Bartik instrument, I use each district's canon share of total budget in 2012 ($s_d^{2012} = \text{canon PIM}_d^{2012}/\text{total PIM}_d^{2012}$), restricting to districts with budgets above 500,000 soles. The mining commodity price index (P_t^{mining}) is a national-level index normalized to 1.0 in the base year, ranging from 0.74 to 1.20 over the sample period.

Financial intermediation. Deposits (*depósitos de ahorro*) and credit (*créditos directos*) at the district-year level come from the SBS statistical appendices. These data cover all regulated financial institutions: commercial banks (*banca múltiple*), municipal savings banks (*cajas municipales*), rural savings banks (*cajas rurales*), finance companies, and the state-owned Banco de la Nación. Districts without any financial institution in a given year are assigned zero deposits and credit.

Financial access. The binary indicator F_{dt} equals one if district d has at least one formal financial office (of any type) in year t. The number of financial offices ($n\_oficinas_{dt}$) serves as a continuous measure. Financial coverage is low and grows slowly: 25.8 percent of districts have at least one office in 2012, rising to 28.9 percent by 2024 (Table 1).

Mining activity. The indicator M_d equals one for districts with active mining operations, based on the Ministry of Energy and Mines registry. Of the 1,893 districts, 334 (18 percent) have mining activity.

Table 1 reports summary statistics. The typical district spends 15 million soles annually, of which 6.7 million comes from canon revenues. Financial coverage is sparse: median deposits and credit are zero, reflecting the large share of districts without banks. Among districts with financial institutions, mean deposits are 384,000 soles and mean credit is 162,000 soles.

Table 1: Summary Statistics

Variable	N	Mean	SD	Median	Min	Max
Panel A: Fiscal variables						
Total spending (000s soles)	24,304	14,972	47,480	5,686	2	2,530,865
Canon revenues (000s soles)	24,303	6,743	25,673	1,703	0	1,188,949
Canon share (2012, s_d^{2012})[a]	1,837	0.445	0.237	0.429	0.003	0.977
Mining price index (P_t^{mining})	24,590	1.00	0.12	0.94	0.74	1.20
Panel B: Financial variables						
Deposits (000s soles)	24,590	93	2,027	0	0	135,399
Credit (000s soles)	24,590	24	467	0	0	33,514
N financial offices	24,583	2.3	9.0	0	0	125
Has financial institution (F_{dt})	24,583	0.27	0.45	0	0	1
Panel C: District characteristics						
Has active mine (M_d)	24,590	0.18	0.38	0	0	1

[a] Cross-section: districts with total PIM $\geq$ 500,000 soles in 2012.
Notes: District-year panel, 2012–2024. Spending and canon in thousands of soles. Deposits and credit from SBS; districts without financial institutions assigned zero. Financial office counts include all regulated institution types.

4 Empirical Strategy

4.1 Estimating Equation

I estimate the effect of municipal spending on local financial intermediation using the following specification:

$$y_{dt} = \beta \log G_{dt} + \gamma F_{dt} + \mu_d + \lambda_t + \varepsilon_{dt} \tag{1}$$

where y_{dt} is $\log(\text{deposits}_{dt} + 1)$ or $\log(\text{credits}_{dt} + 1)$; $\log G_{dt}$ is log municipal spending; F_{dt} is the binary financial access indicator; and μ_d and λ_t are district and year fixed effects. The coefficient of interest is β, which captures the effect of a one-log-point increase in spending on the log of financial intermediation, conditional on bank presence. Standard errors are clustered at the district level throughout. I report province-clustered standard errors as a robustness check, following the concern in Adão et al. (2019) that shift-share designs may induce cross-regional correlation.

The use of $\log(y+1)$ as the dependent variable follows common practice in the literature but raises issues discussed by Chen and Roth (2024). Because the transformation is not scale-invariant, the magnitude of the coefficient depends on the unit of measurement. I address this explicitly in the robustness section by reporting estimates across multiple units (soles, thousands, millions) and confirming that the sign is robust, while acknowledging that the point estimate's magnitude should be interpreted with caution.

The district fixed effects absorb all time-invariant differences across districts—including geography, institutional capacity, pre-existing financial infrastructure, and historical canon dependence. The year fixed effects absorb nationwide trends in financial development, monetary policy, and macroeconomic conditions. The identifying variation is thus *within-district* over time, driven by changes in municipal spending that are plausibly exogenous to local financial conditions.

I include F_{dt} as a control rather than an endogenous variable because financial institution location decisions are made by regulated banks over medium-term horizons and are unlikely to respond to year-to-year spending shocks. The results below confirm this: instrumented spending has no significant effect on the number of financial offices or on the probability of having any bank ($p = 0.998$).

4.2 Bartik Instrument

The OLS estimation of equation (1) faces an obvious endogeneity concern: municipal spending may respond to local economic conditions that also affect deposits and credit. I instrument log spending with a Bartik (shift-share) variable (Goldsmith-Pinkham et al., 2020):

$$Z_{dt} = s_d^{2012} \times P_t^{mining} \tag{2}$$

where s_d^{2012} is the district's canon share of total budget in 2012 and P_t^{mining} is the national mining commodity price index. The instrument captures the idea that districts more dependent on canon revenues in the pre-period experience larger spending shocks when commodity prices fluctuate.

Following the framework of Goldsmith-Pinkham et al. (2020), the identifying assumption is that the pre-period shares s_d^{2012} are exogenous to subsequent trends in financial intermediation, conditional on district and year fixed effects. The district fixed effects absorb the level of s_d^{2012}, so the instrument exploits the *interaction* of cross-sectional exposure with time-series price variation. The year fixed effects absorb any direct effect of commodity prices on national financial conditions.

4.3 Threats to Identification

Two main concerns warrant discussion.

Direct mining exposure. Commodity price booms may affect local financial markets through mining employment, wages, and supplier linkages (Aragón and Rud, 2013; Allcott and Keniston, 2018), independent of the fiscal channel. If mining booms directly affect deposits or credit, the exclusion restriction is violated. I address this with two strategies: (i) controlling for $M_d \times P_t^{mining}$, which absorbs the direct effect of commodity prices on mining districts; and (ii) restricting the sample to the 1,507 districts without any mining activity, which receive canon transfers through regional redistribution but have no direct mining exposure.

Endogeneity of shares. The pre-period canon share s_d^{2012} is correlated with baseline district characteristics. Districts with higher canon shares tend to have fewer financial institutions, lower nighttime light intensity, and more mining activity (Appendix Table 5). However, these level differences are absorbed by the district fixed effects. The identifying assumption requires only that the shares are uncorrelated with *trends* in financial intermediation, conditional on fixed effects—a weaker condition that is supported by the consistency of results across multiple sample restrictions.

5 Results

5.1 First Stage and Reduced Form

Table 2, Panel A reports the first-stage regression. The Bartik instrument is a strong predictor of municipal spending: a one-unit increase in the instrument raises log spending by 0.839 (s.e. = 0.142, $p < 0.001$). The F-statistic is 34.9, well above the Stock and Yogo (2005) threshold. Financial access (F_{dt}) does not predict spending ($\hat{\gamma} = -0.054$, $p = 0.26$), alleviating concerns that bank presence confounds the first stage. The within-R^2 of the first stage is 0.0033, indicating that while the instrument is strong in a statistical sense, it explains a small fraction of within-district spending variation—as expected given the many determinants of municipal budgets.

Reduced form. The reduced form—regressing outcomes directly on the Bartik instrument—provides the cleanest evidence because it bypasses second-stage modeling assumptions about the relationship between spending and intermediation. The Bartik instrument significantly reduces deposits: the coefficient is -0.610 (s.e. = 0.247, $p = 0.014$). For credits, the reduced-form coefficient is -0.752 (s.e. = 0.394, $p = 0.057$). The financial access control F_{dt} enters strongly in the reduced form as well: 4.574 (0.443) for deposits and 3.527 (0.616) for credits. The within-R^2 of the reduced-form deposits equation is 0.274. The Anderson-Rubin test, which is robust to weak instruments, rejects $\beta = 0$ at $p = 0.014$ with district clustering and $p = 0.043$ with province clustering.

Figure 2 displays the reduced form visually. After residualizing both the Bartik instrument and log deposits on district fixed effects, year fixed effects, and financial access, a clear negative relationship emerges: districts with higher Bartik exposure experience lower deposits.

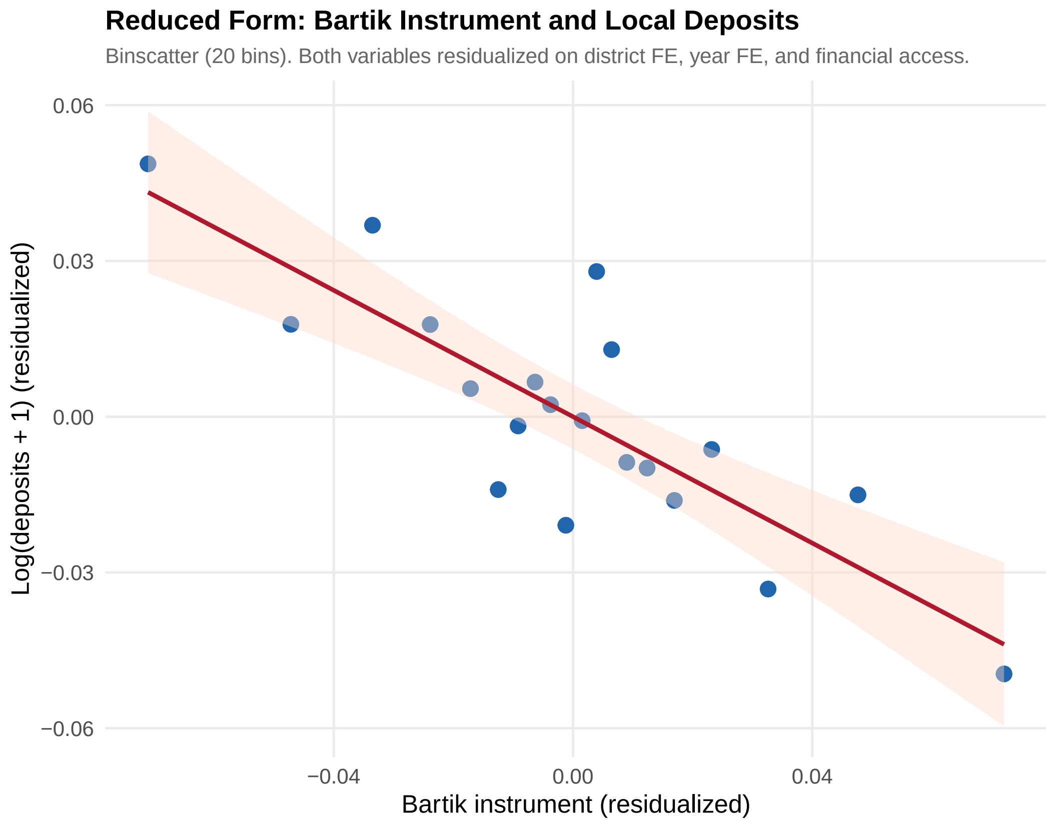

Figure 2: Reduced Form: Bartik Instrument and Local Deposits

Notes: Binscatter with 20 equal-sized bins. Both the Bartik instrument ($s_d^{2012} \times P_t^{mining}$) and $\log(\text{deposits} + 1)$ are residualized on district FE, year FE, and F_{dt}. Line is OLS fit with 95% confidence interval.

5.2 IV Estimates

Table 2, Panel B presents the IV results alongside OLS benchmarks. OLS estimates of β are near zero for both deposits (0.009, s.e. $=$ 0.009) and credit (0.012, s.e. $=$ 0.016), consistent with the expectation that endogeneity biases the OLS estimate upward.

The IV estimates are negative. For deposits—the main outcome—instrumented spending has a coefficient of -0.727 (s.e. $=$ 0.323, $p < 0.05$) using $\log(d+1)$ and -0.744 (s.e. $=$ 0.339,

$p < 0.05$) using the inverse hyperbolic sine. For credits, the IV coefficients are -0.897 (s.e. $= 0.493$, $p < 0.10$) and -0.934 (s.e. $= 0.525$, $p < 0.10$). The sign is robustly negative across all specifications. The credit result provides suggestive evidence of a similar pattern, though it is less precisely estimated and more sensitive to specification choices (as shown in the robustness section).

The coefficient on F_{dt} is large and stable across all specifications: 4.535 (0.447) for deposits and 3.478 (0.621) for credits, reflecting the fact that bank presence is the primary determinant of local intermediation levels.

Table 2: Municipal Spending and Local Financial Intermediation

	Deposits			Credits		
	OLS $\log(d{+}1)$ (1)	IV $\log(d{+}1)$ (2)	IV $\mathrm{asinh}(d)$ (3)	OLS $\log(c{+}1)$ (4)	IV $\log(c{+}1)$ (5)	IV $\mathrm{asinh}(c)$ (6)
Panel A: First stage (dep. var.: $\log G$)						
Bartik (Z_{dt})		0.839^{***} (0.142)			$F = 34.9$	
F_{dt}		-0.054 (0.047)			$p = 0.26$	
Panel B: Reduced form (dep. var.: y_{dt})						
Bartik (Z_{dt})		-0.610^{**} (0.247)			-0.752^{*} (0.394)	
Panel C: OLS and IV estimates						
$\log G$	0.009	-0.727^{**}	-0.744^{**}	0.012	-0.897^{*}	-0.934^{*}
	(0.009)	(0.323)	(0.339)	(0.016)	(0.493)	(0.525)
F_{dt}	4.459^{***}	4.535^{***}	5.022^{***}	3.434^{***}	3.478^{***}	3.754^{***}
	(0.440)	(0.447)	(0.485)	(0.604)	(0.621)	(0.667)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	24,304	23,862	23,862	24,304	23,862	23,862
Within $R^{2\mathrm{b}}$	0.274	0.274	—	0.074	0.074	—

[b] Within R^2 from the reduced-form regression of y_{dt} on the Bartik instrument, F_{dt}, and fixed effects.

Notes: Standard errors clustered by district in parentheses. ***, **, * denote significance at 1%, 5%, and 10%. $\log G$ is instrumented with the Bartik IV in columns 2–3 and 5–6.

5.3 Exclusion Restriction: Fiscal vs. Mining Channels

Table 3 addresses the concern that the Bartik instrument may capture direct mining effects rather than the fiscal channel.

Controlling for direct mining exposure. Columns 1 and 4 add $M_d \times P_t^{mining}$ to the second stage, absorbing any direct effect of commodity prices on mining districts. The spending coefficient is virtually unchanged for deposits (-0.794, s.e. $= 0.325$, $p = 0.015$) and credits (-0.954, s.e. $= 0.494$, $p = 0.054$). Notably, direct mining exposure has a *positive* effect on deposits ($+0.337$, s.e. $= 0.147$, $p < 0.05$), while its effect on credit is positive but insignificant ($+0.289$, s.e. $= 0.246$, $p = 0.24$). This contrast is informative: mining activity generates local deposits—presumably through wages, supplier payments, and economic activity that circulates through local banks—but the fiscal intermediation of mining revenues does not.

Non-mining districts. Columns 2 and 5 restrict the sample to the 1,507 districts without any mining activity ($N = 19{,}572$). These districts receive canon transfers through the regional redistribution formula but have no direct mining exposure. The first stage remains strong ($F = 26.5$). The spending coefficient on deposits is -0.812 (s.e. $= 0.372$, $p = 0.029$). For credits, the point estimate is negative (-0.700, s.e. $= 0.529$) but imprecise.

Post-2017 subsample. Columns 3 and 6 restrict the panel to 2017–2024, a period of commodity price recovery that generates strong within-district spending variation. The first stage is stronger ($F = 47.6$), and both outcomes are significant at the 5 percent level: -0.448 (0.204) for deposits and -0.685 (0.296) for credits.

Figure 3 summarizes the contrast between the fiscal and mining channels. All fiscal spending coefficients lie to the left of zero across both outcomes and all specifications, while the direct mining exposure coefficient is positive.

Table 3: Separating Fiscal from Mining Channels

	Deposits: $\log(d{+}1)$			Credits: $\log(c{+}1)$		
	+ Mine ctl	No mines	Post-2017	+ Mine ctl	No mines	Post-2017
	(1)	(2)	(3)	(4)	(5)	(6)
$\log G$	-0.794^{**}	-0.812^{**}	-0.448^{**}	-0.954^{*}	-0.700	-0.685^{**}
	(0.325)	(0.372)	(0.204)	(0.494)	(0.529)	(0.296)
F_{dt}	4.534^{***}	4.430^{***}	3.993^{***}	3.478^{***}	3.369^{***}	3.028^{***}
	(0.447)	(0.518)	(0.624)	(0.621)	(0.697)	(0.822)
$M_d \times P_t$	0.337^{**}			0.289		
	(0.147)			(0.246)		
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
1st-stage F	36.0	26.5	47.6	36.0	26.5	47.6
N	23,862	19,572	14,680	23,862	19,572	14,680

Notes: SE clustered by district. ***, **, *: 1%, 5%, 10%. All columns instrument $\log G$ with the Bartik IV. $M_d \times P_t$: mine indicator × mining price index. "No mines": districts without mining activity. "Post-2017": 2017–2024 subsample.

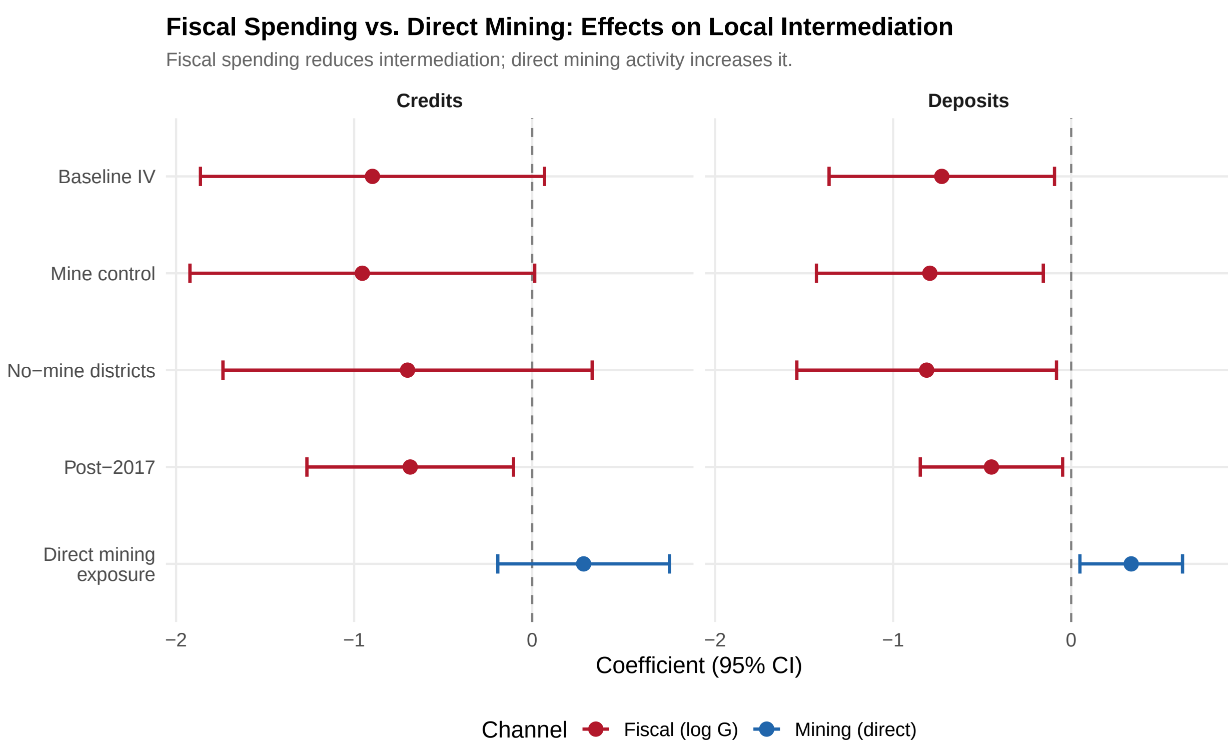

Figure 3: Fiscal Spending vs. Direct Mining: Effects on Local Intermediation

Notes: Point estimates and 95% confidence intervals from Tables 2 and 3. Red: coefficient on instrumented $\log G$ (fiscal channel). Blue: coefficient on $M_d \times P_t^{mining}$ (direct mining channel). All specifications include district and year FE with district-clustered SE.

5.4 Robustness

I organize the robustness checks into three groups: (i) sample and specification choices that might affect statistical precision, (ii) functional-form concerns specific to the $\log(y+1)$ outcome, and (iii) tests that speak to the mechanism and temporal interpretation of the result. Table 4 reports the main specifications; additional results are discussed in the text.

Sample and specification. The deposit coefficient is negative and significant across all sample restrictions and specification choices in Table 4. Clustering standard errors at the province level (196 clusters) rather than the district level inflates them: the deposit p-value rises to 0.12 and the credit p-value to 0.24 (columns 1, 5), and the first-stage F falls to 12.3. While this limits precision at coarser clustering levels, the sign is uniformly negative. Trimming the top and bottom 5 percent of s_d^{2012} yields -0.835 ($p < 0.05$; column 2) for deposits, confirming the result is not driven by outlier shares. Dropping the Lima metropolitan area preserves the negative sign (-0.547, $p = 0.09$; column 3). A potential confound is Peru's *Foncomun*—the largest non-mining transfer, distributed based on welfare indicators—which is correlated with both canon receipts and local economic conditions. Adding $\log(\text{Foncomun}_{dt} + 1)$ as a control leaves the deposit coefficient virtually unchanged at -0.743 (s.e. $= 0.324$, $p = 0.022$; column 4), and raises the first-stage F to 36.6. Foncomun itself enters positively, consistent with its role in financing services in poorer districts, but does not confound the canon result.

Functional form. Three checks address concerns about the $\log(y + 1)$ transformation. First, omitting F_{dt} from the specification yields a deposit coefficient of -0.550 (s.e. $= 0.347$, $p = 0.11$)—the sign is preserved, with some loss of precision, confirming that bank presence does not act as a bad control. Second, as Chen and Roth (2024) note, $\log(y+1)$ coefficients are not scale-invariant. Re-estimating across units gives: $\log(\text{soles}+1)$: -0.727 ($p = 0.025$); $\log(\text{thousands}+1)$: -0.571 ($p = 0.003$); $\log(\text{millions}+1)$: -0.190 ($p = 0.0002$); asinh(soles): -0.744 ($p = 0.028$). The sign is robustly negative and significant in all cases; I emphasize sign and significance rather than a specific elasticity. Third, a PPML control-function approach

(Santos Silva and Tenreyro, 2006) is uninformative in this setting: the high-dimensional fixed effects drop 74% of observations as singletons, reducing the sample to 6,331 district-years. With standard errors of 3.26 and 5.39 for deposits and credits respectively, the PPML estimates carry no information.

Mechanism and dynamics. Two checks address what the result reflects. First, instrumented spending has no significant effect on the probability of having any deposits ($p = 0.31$), any credit ($p = 0.27$), the number of financial offices ($p = 0.33$), or the probability of having any financial institution ($p = 0.998$). The effect therefore operates through the *volume* of intermediation in existing institutions, not through changes in bank presence—consistent with the fiscal leakage mechanism. Second, Figure 4 plots coefficients from interacting s_d^{2012} with year dummies. The 2013 coefficient is near zero ($+0.048$, $p = 0.31$), and from 2014 onward the coefficients become increasingly negative, reaching -0.41 ($p = 0.001$) by 2021. This monotonic decline is consistent with cumulative divergence between high- and low-canon districts. However, since the coefficients do not reverse after 2017 when commodity prices recovered, they may also reflect a differential trend rather than a purely contemporaneous effect. This pattern warrants caution in interpreting the IV estimates as solely driven by annual spending shocks.

Without financial access control. Omitting F_{dt} from the specification yields a deposit coefficient of -0.550 (s.e. $= 0.347$, $p = 0.11$) and a credit coefficient of -0.761 (s.e. $= 0.498$, $p = 0.13$), with $F = 34.8$. The sign remains negative, though precision decreases as expected when the most powerful predictor of intermediation levels is excluded from the regression.

Unit sensitivity. As Chen and Roth (2024) emphasize, the $\log(y + 1)$ transformation is not scale-invariant: the coefficient magnitude depends on the unit of measurement. To assess this concern, I re-estimate the main IV specification measuring deposits in soles, thousands of soles, and millions of soles. The results are: log(soles + 1): -0.727 (0.323, $p = 0.025$); log(thousands + 1): -0.571 (0.189, $p = 0.003$); log(millions + 1): -0.190 (0.051, $p = 0.0002$); asinh(soles): -0.744 (0.339, $p = 0.028$). The sign is robustly negative and

Table 4: Robustness Checks

	Deposits: log(d+1)				Credits: log(c+1)		
	Prov. SE	Trimmed	Excl. Lima	+Foncomun	Prov. SE	Trimmed	Excl. Lima
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\log G$	-0.727	-0.835^{**}	-0.547^{*}	-0.743^{**}	-0.897	-0.272	-0.329
	(0.460)	(0.366)	(0.324)	(0.324)	(0.765)	(0.535)	(0.441)
F_{dt}	4.535^{***}	4.467^{***}	4.615^{***}	4.539^{***}	3.478^{***}	3.364^{***}	3.556^{***}
	(0.462)	(0.454)	(0.448)	(0.445)	(0.742)	(0.651)	(0.629)
log(Foncomun)				0.601^{***}			
				(0.177)			
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1st-stage F	12.3	21.6	32.4	36.6	12.3	21.6	32.4
N	23,862	21,496	21,639	23,822	23,862	21,496	21,639

Notes: All columns instrument $\log G$ with the Bartik IV. Columns 1, 5: SE clustered at province level (196 clusters). Columns 2, 6: excluding top/bottom 5% of s_d^{2012}. Columns 3, 7: excluding Lima. Column 4: adds $\log(\text{Foncomun}_{dt} + 1)$ as control; district-clustered SE.

statistically significant across all units. The magnitude varies with the unit, as expected under the $\log(y + 1)$ transformation. I emphasize the sign and significance rather than a specific elasticity interpretation.

Poisson pseudo-maximum likelihood. As a further check on functional form, I estimate a PPML model with a control-function approach for endogeneity (Santos Silva and Tenreyro, 2006). PPML is designed for count/non-negative data and avoids the $\log(y + 1)$ transformation entirely. However, in this application PPML is severely underpowered: the high-dimensional fixed effects result in 17,531 singleton observations being dropped (74% of the sample), leaving only 6,331 observations for deposits and 4,628 for credits. The control-function coefficient on log spending is +3.25 (s.e. = 3.26, $p = 0.32$) for deposits and -7.67 (s.e. = 5.39, $p = 0.16$) for credits—both statistically insignificant with very large standard errors. The PPML reduced form (Bartik $\rightarrow$ deposits) is similarly insignificant: +3.03 (2.84, $p = 0.29$). Given that three-quarters of observations are dropped as singletons—precisely the districts with zero financial activity that constitute the majority of the sample—I interpret the PPML results as uninformative rather than contradictory. The log-linear specification, which retains the full sample, remains the preferred approach.

Extensive margin and bank presence. Instrumented spending has no significant effect on the probability of having any deposits (-0.031, s.e. $= 0.030$, $p = 0.31$) or any credit (-0.053, s.e. $= 0.048$, $p = 0.27$). Similarly, it has no effect on the number of financial offices ($+0.039$, s.e. $= 0.040$, $p = 0.33$) or on the probability of having any financial institution ($+0.0001$, s.e. $= 0.045$, $p = 0.998$). Instrumented spending neither creates nor destroys bank branches. The negative effect on deposits therefore operates through the *volume* of intermediation in existing institutions, not through changes in bank presence.

Dynamic specification. Figure 4 reports coefficients from interacting the canon share s_d^{2012} with year dummies (base year = 2012), controlling for district and year fixed effects and F_{dt}. The 2013 coefficient is near zero ($+0.048$, $p = 0.31$), providing no evidence of an immediate pre-trend. From 2014 onward, the coefficients become increasingly negative, reaching -0.41 ($p = 0.001$) by 2021 and stabilizing thereafter. The monotonic pattern is consistent with a cumulative divergence in financial intermediation between high- and low-canon-share districts. However, the coefficients do not track the commodity price cycle—they decline continuously even as prices recover after 2017—which could reflect either a persistent causal effect of sustained fiscal exposure or a differential trend correlated with the shares. This pattern warrants caution in interpreting the main results as purely driven by contemporaneous spending shocks.

6 Discussion

6.1 Interpretation

The finding that fiscal transfers reduce local financial intermediation—while mining activity increases it—suggests that the mechanism lies in *how* resources enter the local economy, not how much.

When a mining company operates in a district, it pays wages to workers, purchases

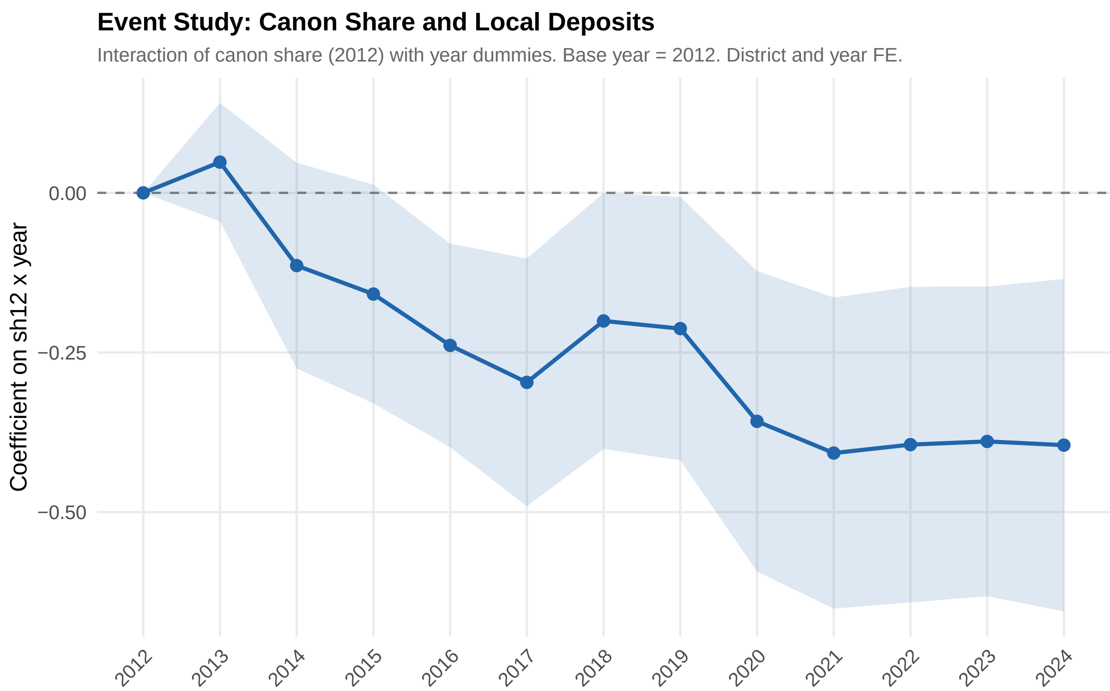

Figure 4: Dynamic Relationship Between Canon Share and Local Deposits

Notes: Coefficients on $s_d^{2012} \times \mathbb{1}[\text{year} = t]$ interactions with 95% CI. Base year = 2012. All specifications include district FE, year FE, and F_{dt}. Standard errors clustered by district.

supplies from local vendors, and generates economic activity that flows through local banks. Workers deposit salaries; businesses take out credit; the banking system deepens. The positive coefficient on $M_d \times P_t$ in Table 3 (+0.337 for deposits) captures this private-sector channel.

When the same mining revenues are intermediated through the fiscal system, the dynamics are different. Canon transfers enter the municipal budget, are processed through the national treasury system (SIAF), and are spent on public investment—roads, schools, water infrastructure. Payments to contractors often flow through the SIAF's centralized payment system rather than through local bank accounts. Contractors may be based outside the district, so their wages and supplier payments circulate elsewhere. The net effect is that fiscal resources enter and leave the district without passing through the local banking system.

This interpretation is consistent with prior findings in the resource curse literature. Caselli and Michaels (2013) show that oil windfalls in Brazilian municipalities increase reported spending but not actual public good provision. Brollo et al. (2013) document that larger transfers attract lower-quality politicians and increase corruption. In Peru, Maldonado and Ardanaz (2023) find that the efficiency of canon spending varies substantially across municipalities, with many districts failing to execute their budgets fully.

A complementary dimension is documented by Contreras and Orihuela (2026), who show that canon transfers reduce municipal property tax collection in Peru over the same period. This matters for the financial intermediation channel: municipalities that collect less in own-source revenue have fewer incentives to maintain banking relationships, as tax collection itself requires bank accounts, payment systems, and financial infrastructure. If fiscal windfalls simultaneously reduce property tax effort and the volume of local intermediation, the two phenomena may reinforce each other—fiscal abundance eroding both the motivation to build revenue institutions and the financial ecosystem that supports them.

The negative coefficient could also reflect a form of Dutch disease at the local level (Allcott and Keniston, 2018): if canon-funded public works crowd out private economic activity that would otherwise generate deposits and credit, the banking system contracts as a byproduct. This is not a pure crowding-out of financial intermediation per se, but rather a compositional shift in local economic activity away from bank-intermediated private transactions toward treasury-intermediated public spending.

6.2 Limitations

Several limitations deserve acknowledgment. First, the outcome variable log(deposits + 1) combines extensive and intensive margins and is not scale-invariant (Chen and Roth, 2024). While the sign is robust across all functional forms and units, the magnitude of the coefficient should not be interpreted as a precise elasticity. The reduced-form coefficient (-0.610, $p = 0.014$) provides the most model-free evidence of the negative relationship.

Second, while province-clustered standard errors inflate confidence intervals beyond conventional significance levels for the deposit result ($p = 0.12$), and the credit result is fragile across specifications, the sign of the coefficient is uniformly negative across all specifications. The deposit result survives all robustness checks except province clustering.

Third, I cannot definitively distinguish between fiscal leakage (money leaves the district), crowding out (public spending displaces private activity), and institutional displacement (fiscal abundance reduces incentives to develop banking). The data are consistent with all three channels, and the true mechanism likely involves some combination.

Fourth, the PPML approach—which would handle zeros more naturally—is uninformative in this setting because the high-dimensional fixed effects eliminate most observations. Alternative approaches to the zeros problem, such as the estimators proposed in Chen and Roth (2024), may provide sharper identification in future work with richer data.

7 Conclusion

This paper provides evidence that fiscal transfers from mining revenues do not generate local financial intermediation in Peru. Using a Bartik instrument that exploits exogenous variation in municipal spending driven by commodity prices, I find that the reduced-form effect of fiscal exposure on deposits is negative and significant ($p = 0.014$), and this negative sign is robust across all IV specifications, functional forms, and sample restrictions. There is also suggestive evidence of a negative effect on credit, though this result is less robust. The negative effect is concentrated in the fiscal channel: controlling for direct mining exposure, or restricting the sample to districts without mines, preserves or strengthens the result. Meanwhile, direct mining activity has a positive effect on deposits, confirming that private-sector resource activity does circulate through local banks.

The contrast between the fiscal and private-sector channels may have implications for the design of intergovernmental transfers in resource-rich developing countries. If fiscal transfers systematically bypass local financial institutions, then complementary policies—such as channeling portions of public spending through local banks or supporting financial infrastructure in transfer-dependent regions—could potentially improve the effectiveness of decentralized spending. However, testing such policy prescriptions is beyond the scope of this paper. A companion study (Author, 2026) shows that financial access moderates the local resource curse, suggesting that the financial channel documented here may have real consequences for local development.

More broadly, these findings suggest that the resource curse may have a financial dimension that operates through the fiscal system. Understanding the conditions under which fiscal transfers build or bypass local financial infrastructure is an important direction for future research.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude (Anthropic) to assist with drafting, editing, and reviewing manuscript text, as well as with coding and robustness checks. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT Authorship Contribution Statement

[**Author**]**:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The panel dataset used in this study was constructed from publicly available administrative sources: municipal fiscal data from the SIAF-MEF Consulta Amigable portal, financial data from the SBS statistical appendices, and a mining commodity price index derived from World Bank commodity price data. A replication package including all analysis code is available at [repository link to be provided upon acceptance].

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Appendix

Table 5: Balance: Correlation of Canon Share (s_d^{2012}) with Baseline Characteristics

Dep. variable (2012 level)	Coefficient	SE	p-value	R^2
Has financial institution (F_{dt})	−0.191	0.043	0.000	0.011
N financial offices	−6.539	0.839	0.000	0.032
Log municipal spending	0.361	0.122	0.003	0.005
Deposits (000s soles)	−127.4	21.4	0.000	0.019
Credit (000s soles)	−57.1	26.6	0.032	0.003
Has active mine (M_d)	0.168	0.038	0.000	0.011
Nighttime lights (mean)	−3.168	0.425	0.000	0.030

Notes: Each row is a separate OLS regression of the 2012 baseline characteristic on s_d^{2012}. $N = 1{,}837$ districts with valid canon share. Heteroskedasticity-robust standard errors. The correlations are absorbed by district fixed effects in the main specifications.